



Trees 2 (Binary Lifting & Euler Tour)

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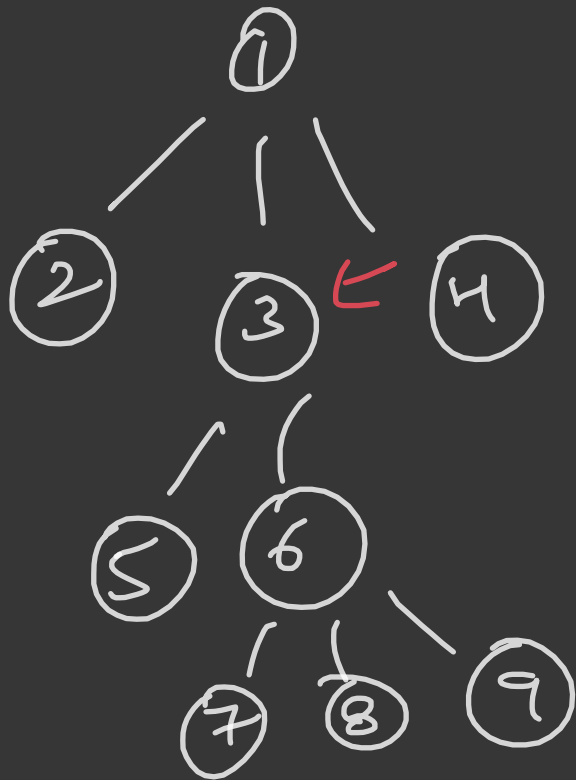


Binary Lifting & Euler Tour

- ✓ Print Kth parent of a node ($n \leq 1e5$)
- ✓ Print Kth parent of a node (query based question $q \leq 1e5$ & $n \leq 1e5$)
- ✓ Binary lifting Intuition
- ✓ Print Kth parent of a node (query based question $q \leq 1e5$ & $n \leq 1e5$) using binary lifting
- ✓ LCA (Lowest Common Ancestor) using Binary Lifting
 - Euler Tour (Tree flattening) - 2 types
 - LCA using Euler Tour

Q. Print the k^{th} parent of a node.

$n \rightarrow 10^5$



$k = 2$

node = 7

output $\rightarrow 3$ ✓

node = 3

parent $\rightarrow$

0	1	1	1	3	3	6	6	6	6
1	2	3	4	5	6	7	8	9	

node = 7

while ($k \neq -1$) {

node = par[node]

}

return node,

$T_c \rightarrow O(n) + O(k) \approx O(n)$

Q q queries given

$q \rightarrow 1e5$

$K \rightarrow n \rightarrow 1e5$

q times $\left[\begin{array}{c} \text{node } K \end{array} \right]$

$$T_c \rightarrow O(n) + O(q \times K) \rightarrow TLE$$

$$\begin{array}{ccc} \downarrow & \downarrow & \uparrow \\ 1e5 & 1e5 & 1e^{10} \end{array} \rightarrow \underline{1e^{10} X}$$

Intuition

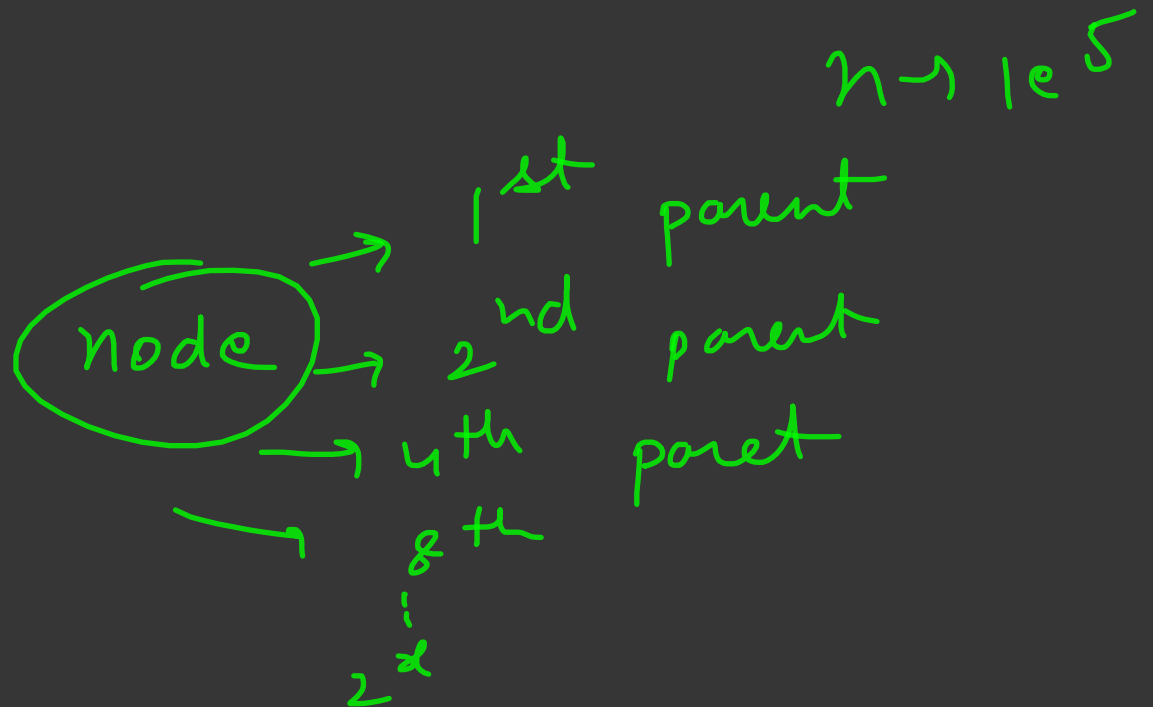
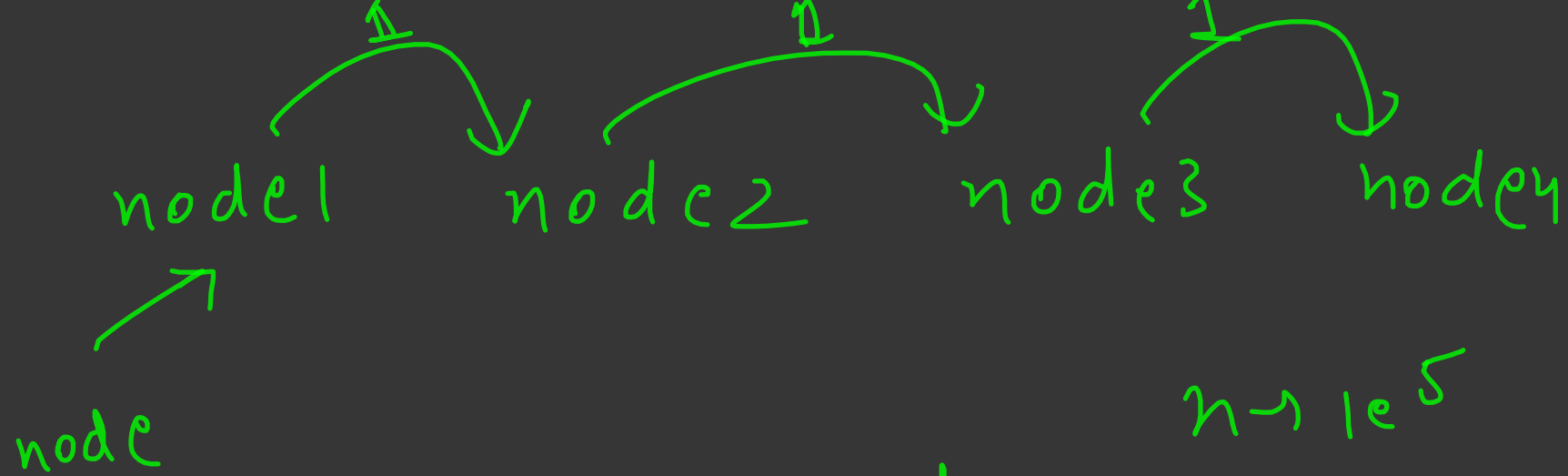
(K) integer

↳ this can be
written as sum of
power of 2.

Eg.

$$K = 11 \rightarrow 1 + 2 + 8 \rightarrow 2^0 + 2^1 + 2^3$$

→ 1011



$$\angle = 11 \rightarrow$$

8 + 2 + 1

↑ ↑ ↑

$$2^0 \rightarrow 1^{\text{st}}$$

21 $\rightarrow 2^{nd}$

$$2^2 \rightarrow 4^{\text{th}}$$

23 → 8th node →

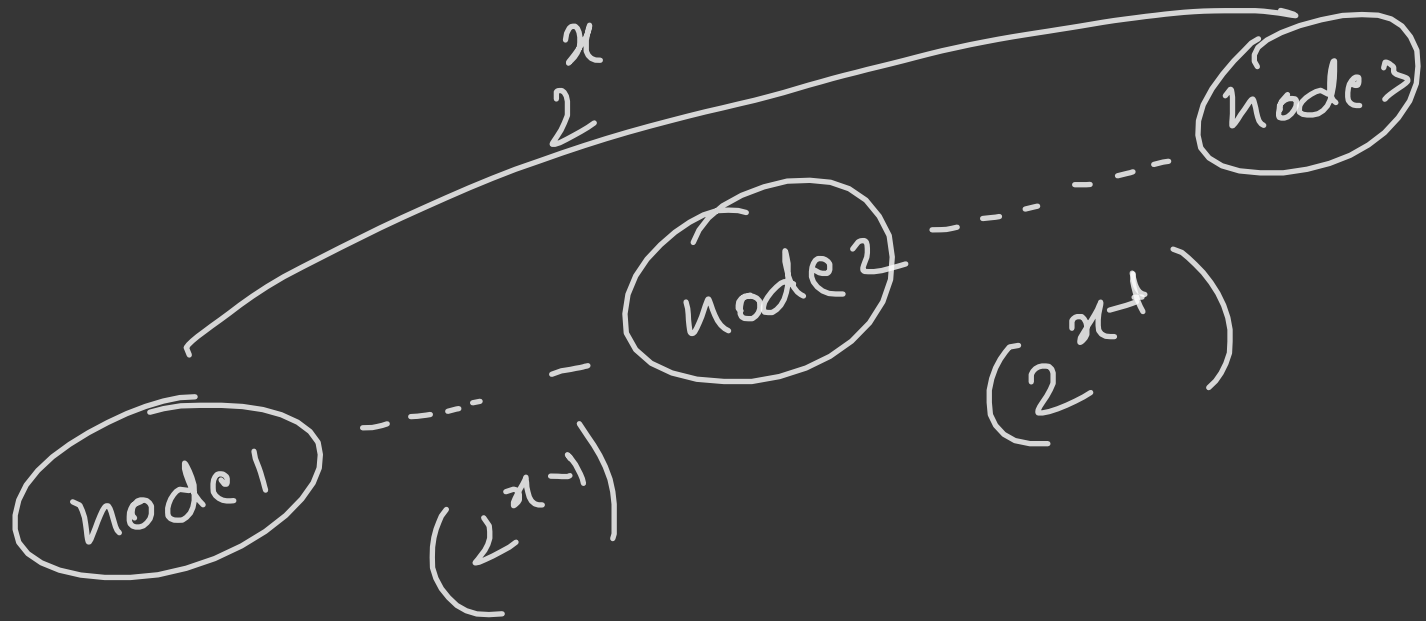
③ $\rightarrow \begin{matrix} 2 & + & 1 \\ 2^1 & & 2^0 \end{matrix}$

$$O(k) \longrightarrow O(\log k)$$

Q how many parents per node?

$\log n$

$2^0 \rightarrow$
 $2^1 \rightarrow$
 $2^2 \rightarrow$
 $2^3 \rightarrow$
⋮



$$\frac{\text{parent}[\text{node1}][x-1] = \text{node2}}{\text{parent}[\text{node2}][x-1] = \text{node3}}$$

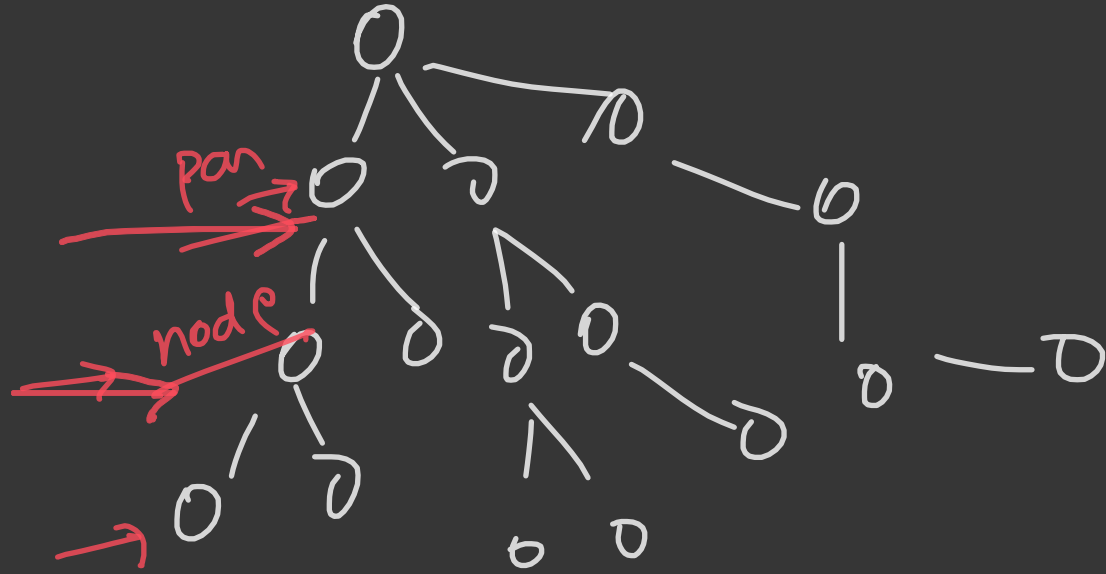
$$\text{parent}[\text{node}][x] = \text{node}3$$

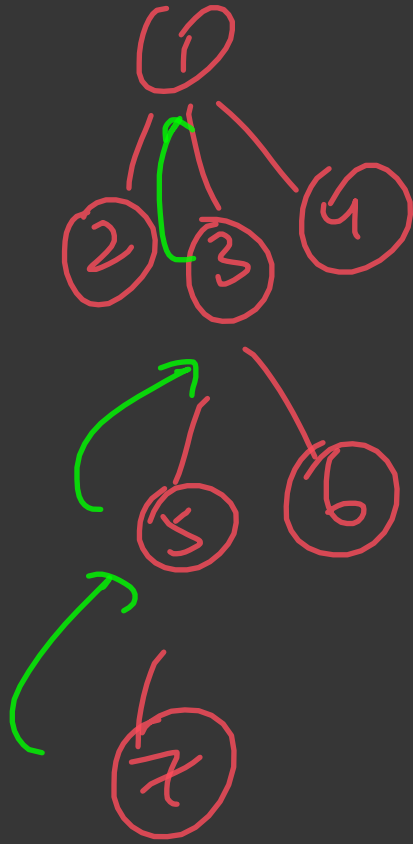
$$\text{parent}[\text{node}1][x] = \text{parent}[\text{node}2][x-1]$$

$$\text{parent}[\text{node}][x] = \text{parent}[\text{parent}(\text{node}][x-1)][x-1]$$

$$\text{parent}[\text{node}][1]$$

$$\text{parent}[\text{node}][0] \rightarrow \checkmark \quad \text{parent}[\text{node}][2]$$





	0	1	2	3	4	...	16
1	0	0	0	0	0	0	
2	1	0	0	0	0	0	
3	1	0	0	0	0	0	
4	1	0	0	0	0	0	
5	3	1	0	0	0	0	
6	3	1	0	0	0	0	
7	5	3	0	0	0	0	

dp

Q Now find k^{th} parent?

cnt = 2

k = 0

```
int cnt = 0,
```

```
while (k) {
```

```
    if (k < 1) {
```

```
        node = dp[node][cnt]
```

```
    }
```

```
    cnt ++;
```

```
    k = k >> 1;
```

node → 1

}
return node,

(11) $\rightarrow$ (101)

$$T_c \rightarrow O(N \log N) + O(q \times \log K)$$

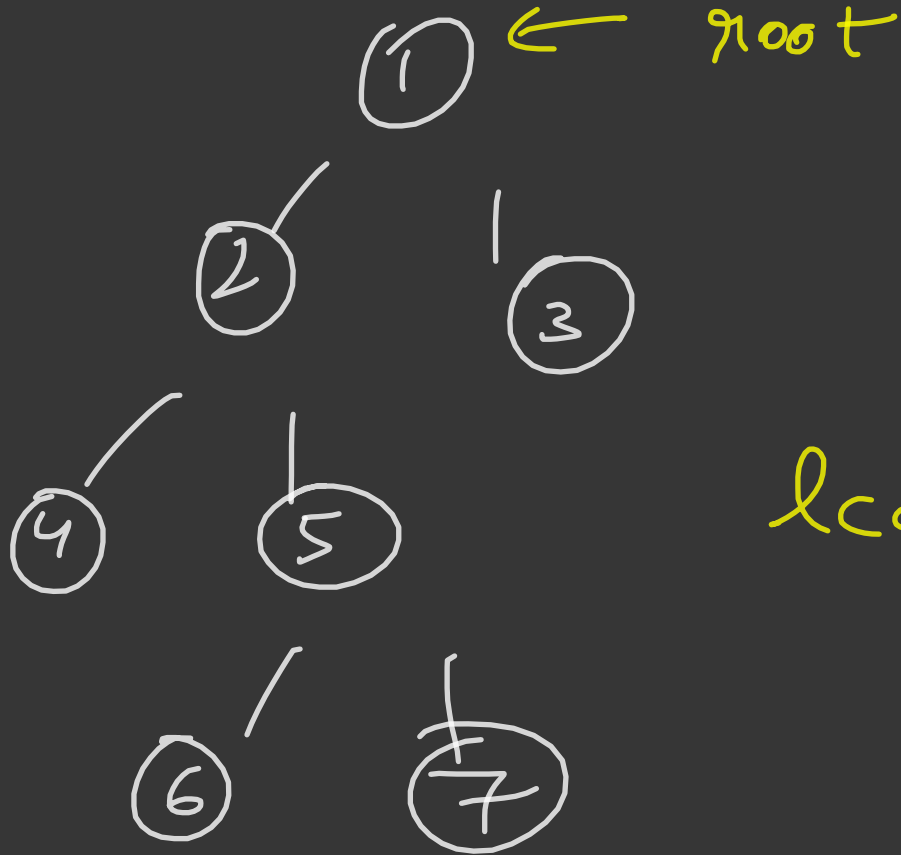
↙
dfs

$\leadsto O(n \log n)$

Q LCA

4 → 2 → 1

5 → 2 → 1

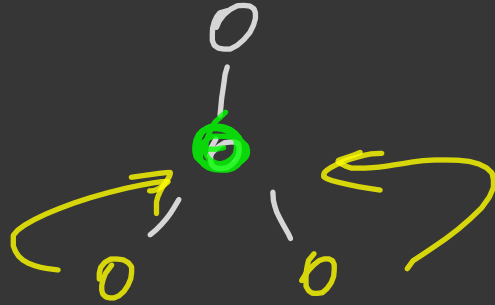


$$lca(\overset{a}{4}, \overset{b}{7}) = 2$$

step 1 → bring them to the same level using binary lifting.

step 2 → Move a & b to every unequal node
* (16 to 0)

$i = 0$



return dp[a][0]

or

dp[b][0]

